

Lie Subgroup and Algebra of Adjoint Image

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User prompt: Let G be a compact (finite-dimensional) Lie group with Lie algebra $\mathfrak{g}$, $Ad : G \rightarrow GL(\mathfrak{g})$ its adjoint representation, and $ad : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$ be the associated Lie algebra homomorphism. Let H be the image of Ad in the general linear group $GL(\mathfrak{g})$. Is it true that the H is a Lie subgroup of $GL(\mathfrak{g})$? If so, is it furthermore true that the Lie algebra of H is the image of ad in $\mathfrak{gl}(\mathfrak{g})$? For each of the two questions, if the answer is affirmative, then give a detailed proof.

Response: Yes to both questions. The image $H = Ad(G)$ is indeed a Lie subgroup of $GL(\mathfrak{g})$, and its Lie algebra is exactly the image $ad(\mathfrak{g})$.

Here are the detailed proofs for both statements.

1. H is a Lie Subgroup of $GL(\mathfrak{g})$

Proof:

To show that H is a Lie subgroup, we rely on **Cartan's Closed Subgroup Theorem**, which states that any topologically closed subgroup of a Lie group is an embedded Lie subgroup.

- Continuity of the Adjoint Representation:** The adjoint representation $Ad : G \rightarrow GL(\mathfrak{g})$ is a smooth map, and therefore continuous.
- Compactness of the Image:** We are given that the Lie group G is compact. Because the continuous image of a compact space is compact, the subgroup $H = Ad(G)$ is a compact subset of $GL(\mathfrak{g})$ under the standard manifold topology.
- Closedness of the Image:** The general linear group $GL(\mathfrak{g})$ is an open subset of the vector space of endomorphisms $End(\mathfrak{g})$, making it a Hausdorff space. In any Hausdorff space, every compact subset is closed. Therefore, H is a closed subgroup of $GL(\mathfrak{g})$.
- Conclusion:** By Cartan's Closed Subgroup Theorem, because H is a closed subgroup of the Lie group $GL(\mathfrak{g})$, H is uniquely equipped with a smooth structure that makes it an embedded Lie subgroup of $GL(\mathfrak{g})$.

2. The Lie Algebra of H is $ad(\mathfrak{g})$

Proof:

Let $\mathfrak{h} \subseteq \mathfrak{gl}(\mathfrak{g})$ denote the Lie algebra of the closed Lie subgroup H . By definition, the Lie algebra of a closed subgroup of a matrix Lie group is given by the set of matrices whose one-parameter subgroups remain entirely within the subgroup:

$$\mathfrak{h} = \{X \in \mathfrak{gl}(\mathfrak{g}) \mid \exp(tX) \in H \text{ for all } t \in \mathbb{R}\}$$

We will prove that $ad(\mathfrak{g}) = \mathfrak{h}$ in two parts: first by showing $ad(\mathfrak{g}) \subseteq \mathfrak{h}$, and second by showing they share the same dimension.

Part A: $ad(\mathfrak{g}) \subseteq \mathfrak{h}$

1. Let $X \in ad(\mathfrak{g})$. By definition, there exists some $Y \in \mathfrak{g}$ such that $X = ad(Y)$.
2. The maps Ad and ad are related by the naturality of the exponential map. Specifically, since $ad = d(Ad)_e$ (the differential of Ad at the identity), the following diagram commutes for all $t \in \mathbb{R}$:

$$\exp_{GL}(t \cdot ad(Y)) = Ad(\exp_G(tY))$$

3. Because $\exp_G(tY) \in G$, its image under Ad must lie in H :

$$Ad(\exp_G(tY)) \in Ad(G) = H$$

4. Therefore, $\exp_{GL}(tX) \in H$ for all $t \in \mathbb{R}$. By the definition of $\mathfrak{h}$, this implies $X \in \mathfrak{h}$. This establishes that $ad(\mathfrak{g})$ is a subspace of $\mathfrak{h}$.

Part B: $\dim(ad(\mathfrak{g})) = \dim(\mathfrak{h})$

1. Let $K = \ker(Ad)$. Since Ad is continuous, K is a closed normal subgroup of G , and therefore a Lie subgroup.
2. The Lie algebra of K , which we will denote as $\mathfrak{k}$, is the kernel of the differential map. Thus, $\mathfrak{k} = \ker(ad) = \{Y \in \mathfrak{g} \mid ad(Y) = 0\}$, which is precisely the center of $\mathfrak{g}$.
3. Because Ad is a Lie group homomorphism with constant rank, it descends to a smooth bijection from the quotient group G/K to H .
4. Since G is compact, the quotient G/K is also compact. A continuous bijection from a compact space to a Hausdorff space (like H) is a homeomorphism. Furthermore, by the constant rank theorem, this map is a diffeomorphism.
5. Because G/K and H are diffeomorphic, their manifold dimensions must be equal:

$$\dim(H) = \dim(G/K) = \dim(G) - \dim(K)$$

6. The dimension of a Lie group is equal to the dimension of its Lie algebra. Therefore:

$$\dim(\mathfrak{h}) = \dim(H) = \dim(G) - \dim(K)$$

7. Now consider the linear map $ad : \mathfrak{g} \longrightarrow \mathfrak{gl}(\mathfrak{g})$. By the rank-nullity theorem from linear algebra:

$$\dim(ad(\mathfrak{g})) = \dim(\mathfrak{g}) - \dim(\ker(ad))$$

8. Substituting $\dim(\mathfrak{g}) = \dim(G)$ and $\dim(\ker(ad)) = \dim(\mathfrak{k}) = \dim(K)$ into the equation yields:

$$\dim(ad(\mathfrak{g})) = \dim(G) - \dim(K) = \dim(\mathfrak{h})$$

Conclusion: Because $ad(\mathfrak{g})$ is a vector subspace of $\mathfrak{h}$ and they have the exact same finite dimension, they must be identical. Thus, the Lie algebra of H is $ad(\mathfrak{g})$.
